

BEC503 — Digital Communication

EC branch, semester 5, 2025 scheme. Phase 1 output, generated by raaghu-pc.

How bits survive a noisy channel — bandpass signals as vectors, coherent and noncoherent modulation, the information-theoretic limit, and the block, cyclic and convolutional codes that approach it.

What is in this folder

Path	Contents
digital-communication.js	The machine-readable content file Phase 2 converts. 5 modules, 80 teaching units, quizzes and assignments.
ppt/	Five .pptx decks, 56 slides each, plus one .inspect.ndjson sidecar per deck.
quizzes/	Five quiz PDFs with answer keys, model answers and marking schemes, plus five Moodle GIFT files.
assignments/	Five assignment PDFs with full four-level rubrics and CO attainment tables.
pyq/	Copies of the branch-level merged question papers and PYQ_Index.pdf.
syllabus/	The syllabus extract, a copy of the source syllabus PDF, and Syllabus_Mapping.pdf.
MANIFEST.json	Counts, provenance and the result of every verification check.

Deck conventions Phase 2 depends on

- Every text shape carries a name: `title`, `objective` or `text`. There are no unnamed text shapes.
- **No PowerPoint animation or slide transition XML is present.** Animation is written as an `Animation instructions:` line in the speaker notes, and Phase 2 rebuilds the motion in React from that line.
- Every slide has a notes block with all six headings in this order: `What to say:`, `Pause cue:`, `Pointing cue:`, `Question to ask students:`, `Animation instructions:`, `Transition style:`, followed by a `[Sources]` line.
- Each unit in the content file carries a `slides` array naming the deck slides that teach it. That array is the join key between the deck and the content file.
- Each unit carries a `visual` id and a `visualSpec`. The `visualSpec` reveal order and the matching slide's `Animation instructions:` line describe the same sequence.

Module index

Module	Title	Hours	Units	Slides	Chapters
1	Bandpass Signals to Equivalent Lowpass and Signalling over AWGN Channels	10	16	56	TB1 (Haykin 2014) Ch. 2.8-2.11 pp. 42-52 and Ch. 7.1-7.4 pp. 323-344
2	Digital Modulation Techniques	10	16	56	TB1 (Haykin 2014) Ch. 7.5-7.8 pp. 344-398 and Ch. 7.10-7.14 pp. 400-418

Module	Title	Hours	Units	Slides	Chapters
3	Information Theory	10	16	56	TB1 (Haykin 2014) Ch. 5.1-5.10 pp. 207-244
4	Error Control Coding: Linear Block and Cyclic Codes	10	16	56	TB1 (Haykin 2014) Ch. 10.1-10.5 pp. 577-605
5	Convolutional Codes	10	16	56	TB1 (Haykin 2014) Ch. 10.6-10.8 pp. 605-623

Sources

- Syllabus: `vtu-3ecesy11-2022.pdf`, VTU EC semester 5, 2025 scheme.
- Prescribed textbook 1: Simon Haykin, *Digital Communication Systems*, John Wiley & Sons, 1st edition, 2014. File: `dokumen.pub_digital-communications-systems.pdf`. Machine-extractable: yes, roughly 2.1 million characters; all TB1 chapter and page references come from the book's own table of contents.
- Previous years' papers: `public/Previous Years QPs/EC/`, copied unchanged. See `pyq/PYQ_Index.pdf` for the scheme-drift warning — Network Analysis is examined under `BEC304` in those papers, not `BEC503`.

Regenerating

```
node temp/.tools/build-decks.mjs <subjectDir> digital-communication.js
node temp/.tools/build-docs.mjs <subjectDir> digital-communication.js <generatedBy> <pyqDir>
<syllabusPdf>
node temp/.tools/verify.mjs <subjectDir> digital-communication.js
```

Generator scripts and `node_modules` live in `temp/.tools/`. Nothing outside `temp/` was written.